Can the video model recognize a revisited place?

Actual Minecraft views from two routes through the same indoor maze





Query: route 1, tick 1472

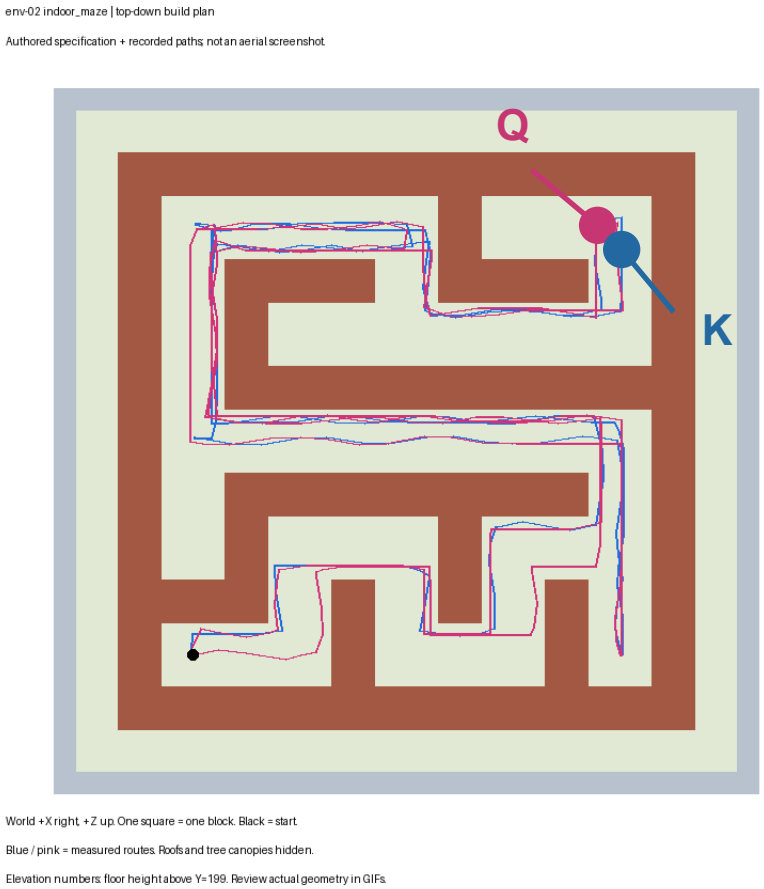
Closest recorded place: route 0, tick 832

These two recorded poses are 1.60 blocks apart, but face 110 degrees apart.

The right image is the coordinate oracle, NOT a claimed model match.

Where are the two views in the world?

A real query and its nearest available route-0 reference



Top-down: authored build + recorded paths, not an aerial image.

Q: where route 1 really was

X 9.50 Y 200 Z 10.59

K: nearest route-0 recording

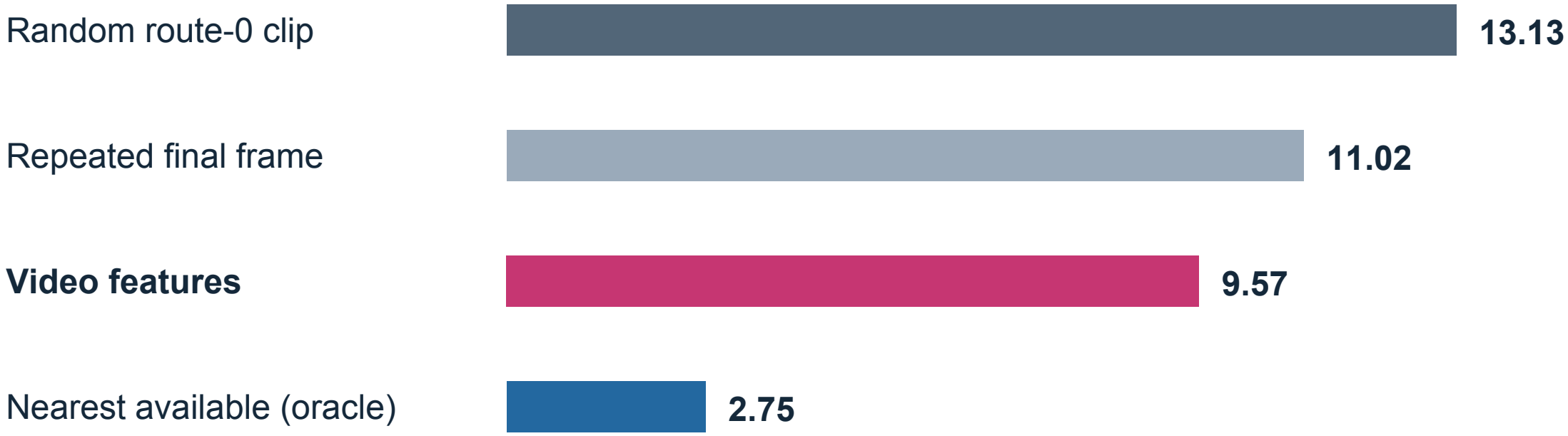
X 10.63 Y 200 Z 9.46

Actual Q-to-K distance: 1.60 blocks

Minecraft coordinates score the match.
They were not inputs to V-JEPA.

What the retrieval test found

Mean 3D distance between each route-1 query and its chosen route-0 clip



Lower is better. Each world contributes equally to the mean.

The model beats random retrieval, but is far from the oracle.

The different-heading test is still small

Query and reference positions within 2 blocks, with at least 90 degrees of heading change

Indoor maze	**3 / 7**	queries matched within 2 blocks
Forest	**0 / 4**	queries matched within 2 blocks
Mine	**5 / 10**	queries matched within 2 blocks

These are correlated clips from only three worlds with enough qualifying revisits.

The report does not establish view-invariant place cells or a cognitive map.

Why test sliding windows next?

More evaluation endpoints from the SAME recordings; frozen model and clip length unchanged



Old: one endpoint per 16-frame block

Proposed: move the 16-frame window by one recorded frame

This can recover missed revisits. Adjacent windows share 15 of 16 frames.

Report independent locations/route pairs, not window count as sample size.

Research precedent: short video sequences for place recognition; sliding temporal windows appear in the literature. A one-frame hop is our evaluation choice, not a published optimum or evidence of biological cells.

Garg & Milford, SeqNet (2021); Zhao et al., RA-L (2024). Sources on final page.

Evidence and next measurement

The 9.57-block result is a world-average error, not one example match.

This JSON did not save the model-selected route-0 clip for each query.

The two real views on slides 1-2 show an oracle pair, never a model prediction.

For a real predicted-vs-actual map figure, the next report must save:

query clip ID and XYZ; selected reference clip ID and XYZ; similarity;
oracle reference ID; heading difference; 3D distance for each query.

Research

Garg & Milford (2021), SeqNet: arxiv.org/abs/2102.11603

Zhao et al. (2024), Sequence Descriptor: arxiv.org/abs/2305.11467

V-JEPA 2 official 16-frame evaluation config: github.com/facebookresearch/vjepa2